

Lecture 14.

Information Retrieval

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Tasks

- Document indexing
- Searching for relevant documents
- Ranking the results

Document indexing

What is an index?

- Index is a pre-built data structure that stores information about the content of documents
- It allows for quickly determining which documents contain specific words or phrases

Why is an index needed?

- Without index, searching would require scanning all documents entirely (can be extremely slow for large collections of data)
- It reduces search time by providing quick access to relevant documents

Main Types of Indexes

- Text Indexes
 - **Representation:** discrete terms (words, phrases), inverted index
 - **Query Types:** exact match
 - **Ranking:** TF-IDF, BM25
- Vector Index
 - **Representation:** continuous vectors (embeddings)
 - **Query Types:** similarity search
 - **Ranking:** cosine similarity, euclidean distance

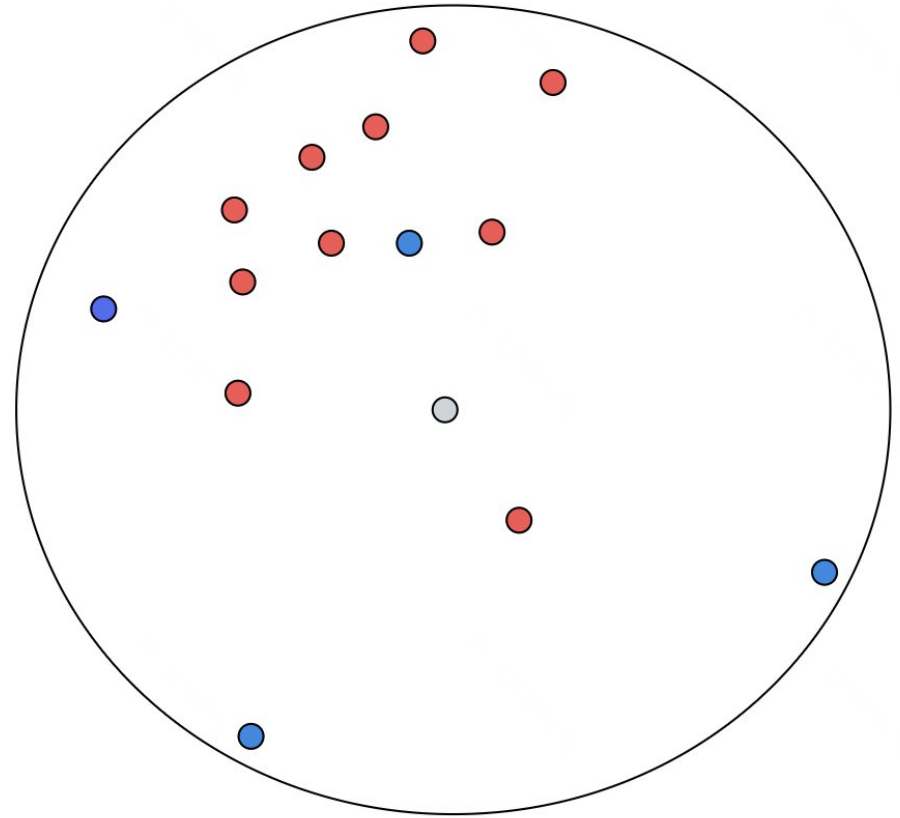
Searching and reranking

Similarity Search Steps

1. Construct an index
2. Encode query
3. Retrieve the top-k most similar items
4. (Optionally) rearrange top-k and choose top-1

Why is reranking needed?

- Suggest that true white point label is blue
- Top-1 prediction is red
- K nearest is also red

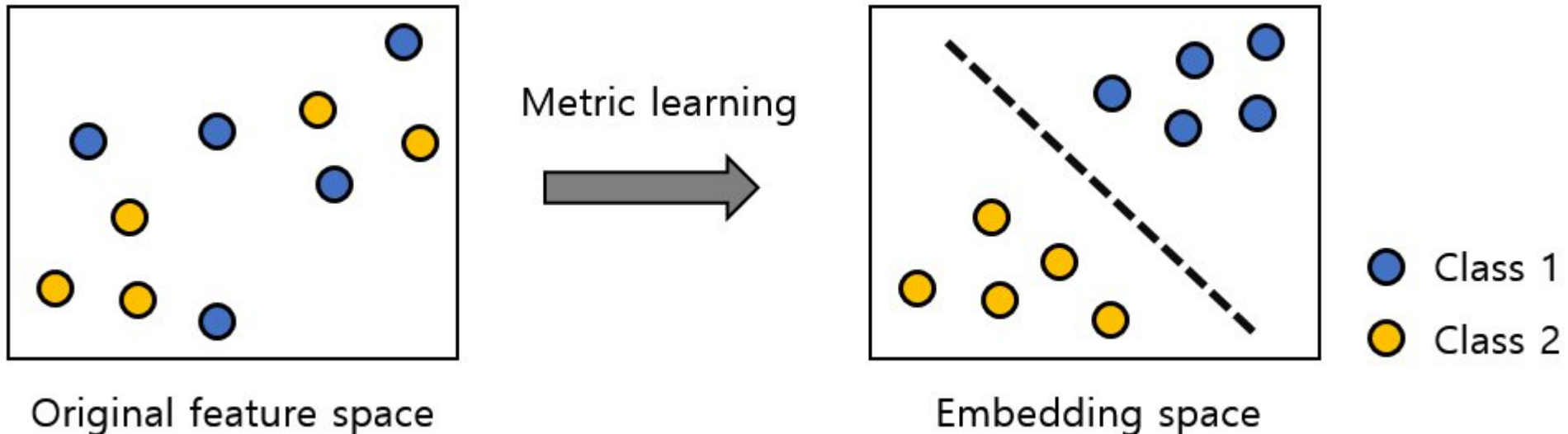


Few-Shot Learning

- FSL is an approach where a model is trained to perform tasks with access to only a small number of training examples for each category
- Examples:
 - Recognizing new objects in images
 - Translating texts for rare languages

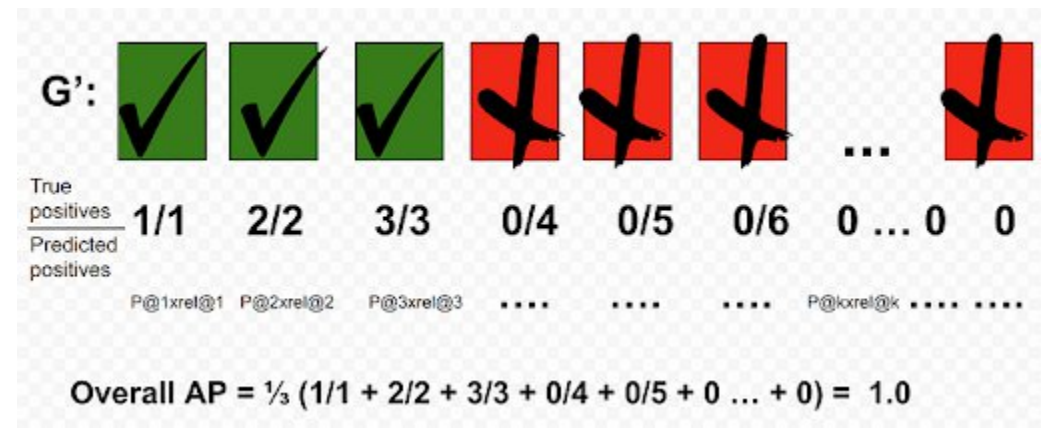
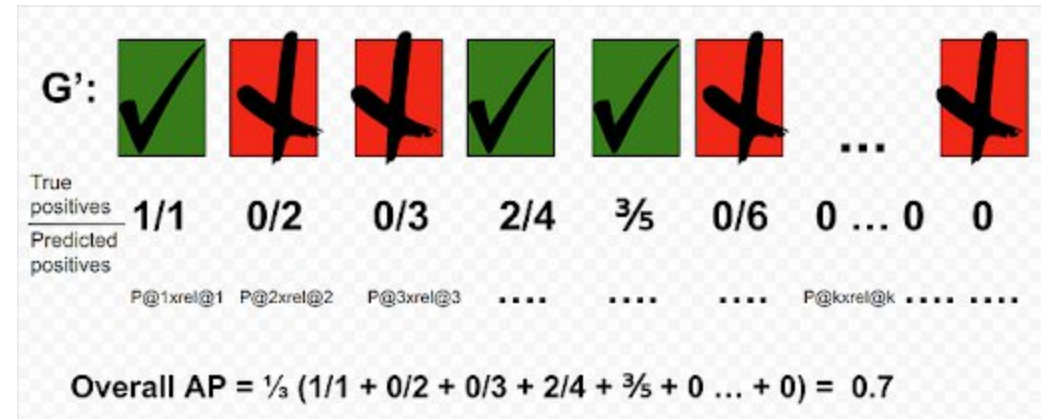
Metric Learning

- In Few-Shot Learning, the key challenge is to generalize across unseen classes using only a few examples
- Metric Learning helps by learning an embedding space where the distance between data points reflect their semantic similarity



Evaluation Metrics

- **Precision at K** is a fraction of relevant results among the top k retrieved documents
- **Recall** as a fraction of the documents that are relevant to the query that are successfully retrieved
- **Average precision** as an area under the precision-recall curve



Extra. RAG in a nutshell

RAG is

- **R** — Retrieval
- **A** — Augmented
- **G** — Generation